

Disproof of conjecture 00000001203

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Verdict: FALSE

We disprove conjecture 00000001203 as stated. The refutation is unconditional and elementary: for the move set $S = \{1, 3\}$ with $k = 3$, the Sprague–Grundy (SG) sequence is $0, 1, 0, 1, \dots$, whose least period is 2. Since $k = 3 \in S$, the conjecture demands period exactly $k + 1 = 4$; it is 2. Moreover $\max(S) = 3 = k$, so the two possible readings of the parameter k coincide and the refutation is interpretation-free. Corroborating witnesses ($S = \{2\}$, $k = 2$, and the clause-1 witness $S = \{1, 2\}$) show that the failure is not caused by our choice of witness.

1 The conjecture

Quoted verbatim from `conjectures/00000001203.md`:

Definition: An addition-subtraction game is a subtraction game whose legal moves are the fixed values in a set S ; the SG period is the least positive period of $g(n)$.

Conjecture: If $S \subset \{1, \dots, k\}$, then the period of g divides $2^k - 1$ if and only if S is nonempty and does not contain k ; when $k \in S$ the period is exactly $k + 1$.

(The conjecture is filed bilingually; the Chinese original is quoted in `README.md`.) The statement has two clauses:

(C1) (*the “if and only if”*) the least period of g divides $2^k - 1 \iff S$ is nonempty and $k \notin S$;

(C2) (*the special case*) if $k \in S$, the least period is exactly $k + 1$.

Both clauses are refuted below. The primary witness fails (C2); the witness $S = \{1, 2\}$ fails (C1) under every reading of k .

2 Subtraction games and the SG recursion

Definition 1 (SG sequence). For a finite set $S \subset \mathbb{N}_{>0}$ of legal moves (an *addition-subtraction game*), the SG sequence $g: \mathbb{N} \rightarrow \mathbb{N}$ is

$$g(0) = 0, \quad g(n) = \text{mex}\{g(n-s) : s \in S, s \leq n\} \quad (n \geq 1),$$

where $\text{mex}(T)$ is the least non-negative integer not in T .

Definition 2 (SG period). The *SG period* is the least positive p with $g(n+p) = g(n)$ for all $n \geq 0$.

3 The primary witness: $S = \{1, 3\}$, $k = 3$

Theorem 3. *Let $S = \{1, 3\}$ and $k = 3$. Then the SG sequence is*

$$g(0), g(1), g(2), g(3), g(4), g(5), \dots = 0, 1, 0, 1, 0, 1, \dots$$

and $g(n) = n \bmod 2$ for all n . Its least period is 2.

Proof. The recursion reads

$$g(0) = 0, \quad g(1) = \text{mex}\{g(0)\}, \quad g(2) = \text{mex}\{g(1)\}, \quad g(n) = \text{mex}\{g(n-3), g(n-1)\} \quad (n \geq 3).$$

We prove $g(n) = n \bmod 2$ by strong induction. The first terms are

$$g(0) = \text{mex} \emptyset = 0, \quad g(1) = \text{mex}\{0\} = 1, \quad g(2) = \text{mex}\{1\} = 0.$$

For $n \geq 3$, both predecessors $n-3$ and $n-1$ have the same parity as $n-1$, so by induction $g(n-3) = g(n-1) = (n-1) \bmod 2 =: a \in \{0, 1\}$. Hence

$$g(n) = \text{mex}\{a, a\} = \text{mex}\{a\} = 1 - a = 1 - ((n-1) \bmod 2) = n \bmod 2.$$

Thus $g(n+2) = g(n)$ for all n , so 2 is a period, while $g(1) = 1 \neq 0 = g(0)$ excludes period 1. Therefore the least period is exactly 2. $\square$

Corollary 4. *The primary witness refutes conjecture 00000001203. Here $k = 3 \in S$, so clause (C2) demands least period $k+1 = 4$; the actual least period is $2 \neq 4$. Equivalently, there is no way for the period to be both 4 and the least period, because 2 already is.*

Remark 5. For this witness clause (C1) happens to be *satisfied*: $k = 3 \in S$ makes the right-hand side false, and $2 \nmid 2^3 - 1 = 7$ makes the left-hand side false as well, so the biconditional is true. It is clause (C2) that fails. This is exactly why we also exhibit a clause-(C1) witness in Section 5.

4 The reading of the parameter k is ambiguous — and the primary witness is immune

The conjecture says “if $S \subset \{1, \dots, k\}$ ”, which admits two natural readings of k :

- (A) *Ambient bound.* k is the bound in $\{1, \dots, k\}$, so the hypothesis is $S \subseteq \{1, \dots, k\}$ for a k that may strictly exceed $\max(S)$. This reading is supported by the wording: “does not contain k ” is a genuine, non-vacuous condition only when k can lie outside S .
- (B) *Charitable / tight reading.* $k = \max(S)$.

Under reading (B) the condition “ $k \notin S$ ” is never satisfiable (because $\max(S) \in S$ by definition whenever $S \neq \emptyset$), so the “if and only if” degenerates. We therefore treat both readings explicitly.

Proposition 6. *The primary witness $S = \{1, 3\}$, $k = 3$ refutes clause (C2) under both readings (A) and (B): under (A) the ambient bound is $k = 3$; under (B) we have $\max(S) = 3$, so again $k = 3$. Since the two values agree and $k \in S$, clause (C2) demands period $k+1 = 4$ in either case, while the least period is 2.*

Thus the disproof does not depend on resolving the ambiguity. By contrast, the earlier candidate $S = \{1\}$ with $k = 3$ is fragile: under reading (B) one has $k = 1$ and the period $2 = k+1$ satisfies clause (C2), so that witness refutes the conjecture only under reading (A). We do not rely on it.

5 Corroborating witnesses

Second-clause witness: $S = \{2\}$, $k = 2$

Theorem 7. *Let $S = \{2\}$. Then*

$$g = 0, 0, 1, 1, 0, 0, 1, 1, \dots, \quad g(n) = \left\lfloor \frac{n}{2} \right\rfloor \bmod 2,$$

with least period 4. Since $k = 2 \in S$, clause (C2) demands least period $k + 1 = 3$; but the least period is $4 \neq 3$. Again $\max(S) = 2 = k$, so the witness is interpretation-free.

Proof. Here $g(n) = \text{mex}\{g(n-2)\}$ for $n \geq 2$, with $g(0) = g(1) = 0$ (for $n = 1$ the only move, 2, is illegal). Since $\text{mex}\{0\} = 1$ and $\text{mex}\{1\} = 0$, the sequence alternates $0, 1, 0, 1, \dots$ starting from $g(2)$ but with the initial double $g(0) = g(1) = 0$; hence $g(n) = \lfloor n/2 \rfloor \bmod 2$, which has least period 4. (The values are $0, 0, 1, 1$ repeating, so no smaller positive period works.) $\square$

First-clause witness: $S = \{1, 2\}$

Theorem 8. *Let $S = \{1, 2\}$. Then $g = 0, 1, 2, 0, 1, 2, \dots$, i.e. $g(n) = n \bmod 3$, and the least period is 3. Moreover $3 \mid 2^2 - 1 = 3$.*

Proof. $g(0) = 0$, $g(1) = \text{mex}\{g(0)\} = 1$, and for $n \geq 2$, $g(n) = \text{mex}\{g(n-2), g(n-1)\}$. The first values are $0, 1, 2, 0, 1, 2, \dots$: from $\{0, 1, 2\}$ the mex of the two preceding distinct residues is the third, and the pattern repeats cyclically. Hence $g(n) = n \bmod 3$ and the least period is 3 (the values $0, 1, 2$ are pairwise distinct, so periods 1 and 2 are impossible). Finally $2^2 - 1 = 3$ and $3 \mid 3$. $\square$

Corollary 9. *Clause (C1) fails for $S = \{1, 2\}$ under both readings of k .*

Charitable reading (B): $k = \max(S) = 2$. Since $k \in S$, the right-hand side “ S is nonempty and $k \notin S$ ” is false; clause (C1) therefore requires the period not to divide $2^k - 1 = 3$. But the period is 3 and $3 \mid 3$, so the required divisibility holds and the biconditional fails.

Ambient reading (A): take $k = 3$; then $S = \{1, 2\}$ is a proper subset of $\{1, 2, 3\}$, the right-hand side is true ($3 \notin S$), so clause (C1) requires $3 \mid 2^3 - 1 = 7$; but $3 \nmid 7$, and the biconditional fails again.

Remark 10 (On the only conceivable escape). One might try to rescue the conjecture by restricting clause (C2) to the full sets $\{1, \dots, k\}$ (the only move sets for which $k = \max(S)$ and $S = \{1, \dots, k\}$ coincide). The text does not state such a restriction, and it would not help anyway: clause (C1) is refuted by $S = \{1, 2\}$ (Corollary 9), which for ambient $k = 3$ is a proper subset and for the charitable reading has $S \subsetneq \{1, \dots, k\}$ as well. We also emphasise that the exact reading of k is genuinely ambiguous in the conjecture text (ambient bound versus $\max(S)$); we have deliberately chosen the primary witness so that the ambiguity is immaterial.

6 Least periods for $k = 2, 3, 4$

Table 1 lists the least period of g for every nonempty $S \subseteq \{1, \dots, k\}$, together with the conjecture’s prediction under both readings of k : reading (A) with ambient bound k , reading (B) with $k = \max(S)$. A “fails” entry means that one of the clauses (C1), (C2) is violated by that row.

k	S	least period	reading (A): ambient k	reading (B): $k = \max(S)$
2	$\{1\}$	2	fails	holds
2	$\{2\}$	4	fails	fails
2	$\{1, 2\}$	3	fails	fails
3	$\{1\}$	2	fails	holds
3	$\{2\}$	4	fails	fails
3	$\{3\}$	6	fails	fails
3	$\{1, 2\}$	3	fails	fails
3	$\{1, 3\}$	2	fails	fails
3	$\{2, 3\}$	5	fails	fails
3	$\{1, 2, 3\}$	4	holds	holds
4	$\{1\}$	2	fails	holds
4	$\{2\}$	4	fails	fails
4	$\{3\}$	6	fails	fails
4	$\{4\}$	8	fails	fails
4	$\{1, 2\}$	3	holds	fails
4	$\{1, 3\}$	2	fails	fails
4	$\{1, 4\}$	5	fails	fails
4	$\{2, 3\}$	5	holds	fails
4	$\{2, 4\}$	6	fails	fails
4	$\{3, 4\}$	7	fails	fails
4	$\{1, 2, 3\}$	4	fails	holds
4	$\{1, 2, 4\}$	3	fails	fails
4	$\{1, 3, 4\}$	7	fails	fails
4	$\{2, 3, 4\}$	6	fails	fails
4	$\{1, 2, 3, 4\}$	5	fails	fails

Table 1: Least periods of the SG sequence for every nonempty $S \subseteq \{1, \dots, k\}$, $k = 2, 3, 4$, and whether conjecture 00000001203 holds for that row under each reading of k . The primary witness is the row $k = 3$, $S = \{1, 3\}$ (period 2 versus the demanded $2k + 1$); the secondary witness is $k = 2$, $S = \{2\}$; the clause-1 witness is $S = \{1, 2\}$. Note that $S = \{1\}$ is the fragile case: it “holds” under reading (B) because then $k = 1$ and the period $2 = k + 1$.

Remark 11. Reading (A) is violated in 3 of 3 rows for $k = 2$, 6 of 7 rows for $k = 3$, and 13 of 15 rows for $k = 4$. Reading (B) is violated in 2, 5, 13 of those rows respectively. In particular each table contains violating rows under either reading, so the conjecture is false no matter how k is read.

7 Reproducibility

The finite computations are carried out by `reproduce.py`, which uses only the Python 3 standard library:

```
$ python3 reproduce.py
... (witnesses, all periods, and the violations tables)
PASS: all checks verified; conjecture 00000001203 is FALSE.
```

It computes g by the mex recursion for all nonempty $S \subseteq \{1, \dots, k\}$ with $k = 2, 3, 4$ over 600 terms and extracts the least period by direct search. It verifies in particular:

- $S = \{1, 3\}$: $g(n) = n \bmod 2$ for the first 600 terms, least period 2, whereas the conjecture (as $3 \in S$) demands $k + 1 = 4$;
- $S = \{2\}$: least period $4 \neq 3 = k + 1$;
- $S = \{1, 2\}$: least period 3 with $3 \mid 2^2 - 1$ (clause (C1) fails under the charitable reading), and $3 \nmid 2^3 - 1 = 7$ for ambient $k = 3$ with S a proper subset (clause (C1) fails under the ambient reading).

It prints `PASS` and exits 0 exactly when every check holds, and exits non-zero otherwise.

8 Formal verification

The refutation is also formalised in core Lean 4 (no Mathlib, no `sorry`) under `lean4/`. The primary witness is the well-founded recursion

```
def g : Nat → Nat
| 0 => 0
| 1 => mex [g 0]
| 2 => mex [g 1]
| n+3 => mex [g (n+2), g n]
```

with the closed form `g_eq : g n = n % 2` proved by `Nat.strongRecOn`, the least-period predicate `IsLeastPeriod`, and the two key theorems

```
theorem least_period_two : IsLeastPeriod 2
theorem not_least_period_four : ¬ IsLeastPeriod (3 + 1)
```

The clause-1 witness is formalised as `g12` with `least_period_three_12 : IsLeastPeriod12 3` and `three_dvd_two_sq_sub_one : 3 ∣ 22 - 1`. All of these are collected in `conjecture_00000001203_false`. The axiom audit lake env lean `Check.lean` reports only `[propext, Quot.sound]` and no `sorryAx`.

9 Status against the submission rules

Rule 3 requires each submission to contain the LaTeX source code, a PDF document, and a Lean 4 project.

- **LaTeX source** — present (`main.tex`), a standalone article that uses only `geometry`, `amsmath`, `amssymb`, `amsthm`, `array`, `parskip`, and `booktabs`, and compiles with `tectonic main.tex`.
- **PDF document** — present at `build/main.pdf`, produced by `tectonic main.tex` and moved into `build/`.
- **Lean 4 project** — present under `lean4/`, pinned to `leanprover/lean4:v4.33.1` with library `tlmc1203`; builds with `lake build`; the audit `lake env lean Check.lean` shows no `sorryAx` and no Mathlib dependency.
- **Reproduction** — `python3 reproduce.py` prints `PASS` and exits 0.